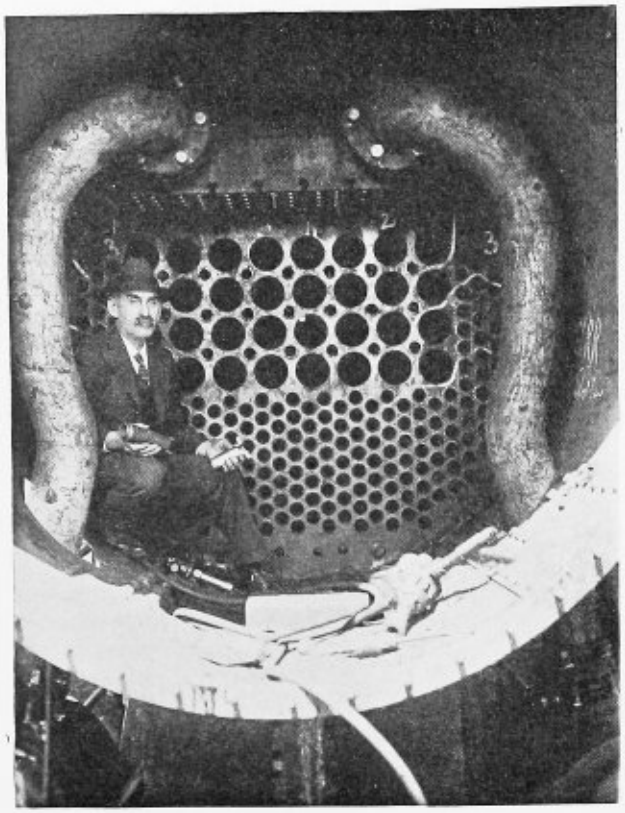


Fig. 1.—Sections of Flue End Ready for Installation

plate about the diameter of the outside of the tube, or slightly larger. The end of the tube is then swedged down about ⅛ inch to permit a copper ring or ferrule to be placed between the tube and the plate. The ferrule for a new firebox is generally 1/16 inch thick. It is the width of the thickness of the tube plate or slightly wider, generally being ⅝ inch. The copper ferrule has two primary functions, i. e., first, it acts as a soft gasket between the tube and the plate and reduces leakage trouble of tubes. Second, the copper ferrule is a greater conductor of heat than the steel tube and tube plate, hence the more rapid flow of heat through the copper ring to the water in the boiler tends to prevent overheating the ends of the tubes. The copper ferrule is fastened in a hole in the tube plate; the tube end is inserted into the ferrule until about 3/16 inch of the tube projects beyond the face of the tube plate on the fire side. The end of the tube is slightly expanded with the mandrel pin, thus fastening the tube in position. After this the tube is prosser-expanded in the plate, the projecting end of the tube beaded over with the beading tool, and the tube lightly rolled with the roller expander.

Use of Expanders

The prosser expander makes a corrugation in the tube next to the water side of the tube plate. The object of this corrugation is to make the tube act as a strut between the front and back tube plates to stiffen the hold of the tube in the back tube plate. This corrugation against the water side of the plate and the bead clinched over on the fire side of the plate firmly fasten the tube in the plate, to the end that the uneven expansion and contraction in the set of flues or tubes cannot move or loosen the hold or joint of the tube in the firebox tube plate. The present method of fastening tubes in boilers is based on this theory.

The result of the use of the prosser expander is a vast improvement over the use of the roller expander only in setting tubes. The semi-circular corrugation on the water side of the plate in the tube greatly assists in holding the tube tight in the tube plate when the tube plate is breathing. It follows that if this semi-circular corrugation with only the edge or beginning of the circle coming in contact with the tube plate forms a strut and assists in holding the tube tight in the plate, how much more will this new device do the work with its sharp shoulder machined in the tube and placed against the water side of the plate.

The flue end obviates all of the conditions that wear out the tubes. With this device there is no use for the prosser expander. There is a mechanically correct shoulder on the tube that does not wear out and require reforming. It is a permanently formed shoulder set against the water side of the plate to remain in its original position holding the tube tightly in place. The roller expander, unlike the prosser expander, draws the tube into the tube plate and tends to enlarge the bead on the fire side of the plate.

This new system also improves the design of the copper ferrule. Instead of using a straight ferrule projecting through the tube plate on the water side ⅛ inch or more to act as a scale trap, or acid former, the ferrule is flanged before it is applied.

In bad feed water districts where the constant use of the prosser expander is necessary to maintain tubes in serviceable condition, with the standard method, the back tube plate is quickly destroyed. The frequent driving of the expander pin into the expanders in the tube has the effect of drifting and stretching the tube holes, with the result that the tube plate is distorted, buckled and holes knocked out of round, the top flange worked up high in the buckle, frequently causing the side flanges as well as the top flange to crack along the knuckles.

The new system obviates all of these evils, making it practical to run the tube plate as long as the rest of the firebox plates in all cases. As mentioned, the use of the prosser expander tapers the holes in the tube plate the wrong way, i. e., large on the water side and small on the fire side, while with the new device the hole in the tube plate is given a slight taper to begin with, the hole in the tube plate being slightly larger on the fire side of the plate. When the end of the tube is set in this hole and the bead clinched over on the fire side of the plate, the holding power of the tube in the plate is something great, because it does not depend on the bead and friction tension of the tube in the hole entirely, but partly on the tube pulling against the taper. An occasional use of the roller expander will maintain this slight taper. At the same time the roller expander tends to draw the tube toward the firebox, thus maintaining the sharp shoulder tight against the tube plate. When beading the tubes, the shoulder on the water side will be drawn up tight much the same as when the head of a rivet is drawn up to the plate on driving.

Resetting Tubes

On removing a set of tubes from a boiler having these new tubes, the tube sheet is in good condition, ready to reapply a set with new points welded on. In applying the tubes all beads would be of uniform size because the tube would be jammed up against the shoulder on the point exactly setting itself in the proper position, whereas it now requires care and good judgment to set the slipping tube exactly in its proper position to allow the right and uniform bead to project beyond the fire side of the plate. Many tubes now leak because some beads are large and some small in the same installation.